MTH2008 – Real Analysis

2026/27

0 – Introduction

Who am I

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- Lecturer in Mathematics
  University of Exeter
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- Knowledge Exchange Champion
  Knowledge Exchange Hub for the Mathematical Sciences
- Advisory board
  Academy for the Mathematical Sciences
- Peer Review College
  Engineering and Physical Sciences Research Council

Motivation

- Real Analysis is where mathematics becomes rigorous.
- Up until now, you have used calculus, limits, and infinite series because they worked.
- Real Analysis asks: *why do they work?* It gives us the solid foundations behind every formula you have learned.
- By the end, you will not only know new theorems, but hopefully, you will see mathematics itself in a sharper, more beautiful light.
- Applications of Real Analysis in my own research includes:
  - image based cancer diagnostics, and
  - ultrasound signal analysis for diagnosing and predicting battery cell defects.
- Other applications include:
  - building LLM's,
  - stochastic processes which are used in finance, AI and data science,
  - signal processing, and
  - thermodynamics, quantum mechanics, and electromagnetism.

Learning outcomes

Module Specific Skills and Knowledge

- State and prove key theorems in real analysis using a rigorous approach.
- Develop proofs related to topological concepts such as limits and connectedness.
- Understand the basis of integration of functions of a real variable.

Discipline Specific Skills and Knowledge

- Apply fundamental mathematical concepts, manipulations and results in analysis.
- Formulate rigorous arguments as part of your mathematical development.

Transferable/Employment Skills and Knowledge

- Think analytically and use logical argument and deduction.
- Communicate your ideas effectively in writing and verbally.
- Manage your time and resources effectively.

Course structure

- There are 3 lecture slots per week:
  - Monday 11:30-12:30
    Amory – Parker Moot Lecture Theatre
  - Tuesday 08:30-09:30
    Streatham Court – Lecture Theatre A
  - Thursdays 15:30-16:30, repeated 16:30-17:30
    Queens – Lecture Theatre 1
- There are 2 Graded Formative Assessments
  These do not count towards your final grade,
  but if you would like feedback on your work, these are to be submitted in Weeks 3 and 8 via ELE.
- There are 5 tutorials happening on the Wednesday of Weeks 2, 4, 7, 9 and 11
  - Tutorial 1 (Week 2): Exercise Sheet 1
  - Tutorial 2 (Week 4): Graded Formative Assessment 1
  - Tutorial 3 (Week 7): Exercise Sheet 2
  - Tutorial 4 (Week 9): Graded Formative Assessment 2
  - Tutorial 5 (Week 11): Exercise Sheet 3
- The assessment for the course will be
  - an on-campus closed book in-person written 2-hour final exam worth 70% of your final grade,
  - 2 Summative Assessment,
    each worth 15% of your final grade
    to take place in-class on 20 Oct. and 24 Nov.
- Office hours
  Tuesdays 09:45-11:15 (Harrsion – Room 205)
  in Weeks 2, 3, 5, 7, 8, 9, 10 and 11.

1 – The Real Numbers

We begin with the study of the real number system.

- § 1.1 deals with the axioms that define the real numbers, definitions based on them, and some basic properties that follow from them.
- § 1.2 introduces ideas of set theory in the context of sets of real numbers. Here, we prove two fundamental results:
  - **Heine–Borel Theorem** and
  - **Bolzano–Weierstrass Theorem**.

1.1 The Real Number System

We denote the set of **real numbers** by the symbol $\mathbb{R}$, on which we have the binary operations of addition $(+)$ and multiplication $(\cdot)$. The real numbers $\mathbb{R}$ equipped with these operations form what is know as a **field**, specifically they satisfy the following operations:

(a) **Commutativeity**
$a + b = b + a$ and $ab = ba$ for all $a, b \in \mathbb{R}$

(b) **Associativeity**
$(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in \mathbb{R}$

(c) **Distributiveity**
$a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in \mathbb{R}$

(d) **Additive identity**
$0 + a = a$ for all $a \in \mathbb{R}$
Multiplicative identity
$a \cdot 1 = a$ for all $a \in \mathbb{R}$

(e) **Additive inverse**
For $a \in \mathbb{R}$ there is a real number $-a$ with $a + (-a) = 0$
Multiplicative inverse
For $a \in \mathbb{R}$ with $a \neq 0$, there is a real number $1/a$ so that $a \cdot (1/a) = 1$.

A set on which two binary operations are defined so as to have properties (a)–(e) is called a **field**. The real number system is by no means the only field.

The order relation

Not all sets have an ordering. For instance think about the set comprising of an orange, an apple and a pear – there is no natural ordering on this three element set.

However, the real numbers are ordered by the relations

$$< \quad > \quad \leq \quad \geq .$$

Indeed, in addition to the field properties, we have:

(f) For each pair of real numbers a and b, exactly one of the following is true

$$a = b, \quad a < b, \quad \text{or} \quad b < a.$$

(g) If $a < b$ and $b < c$, then $a < c$

(h) If $a < b$, then $a + c < b + c$ for $c \in \mathbb{R}$

If $a < b$, then $a \cdot c < b \cdot c$ for $c \in \mathbb{R}$ with $c > 0$

A field with an order relation satisfying (f)–(h) is called an **ordered field**.

Thus, the real numbers together with the binary operations of addition and multiplication, and the order relation $<$ form an ordered field.

Theorem (The triangle inequality)
If a and b are any two real numbers, then

$$|a+b| \leq |a| + |b|.$$

Corollary (The reverse triangle inequality)
If a and b are any two real numbers, then

$$|a-b| \geq ||a| - |b|| \quad \text{and} \quad |a+b| \geq ||a| - |b|| \,.$$

Supremum of a set

A set S of real numbers is **bounded above** if there is a real number b such that $x \leq b$ whenever $x \in S$.

In this case, b is called an **upper bound** of S.

If b is an upper bound of S, then so is any larger number because of the property of transitivity of our order relation $<$.

If β is an upper bound of S, but no number less than β is, then β is a **supremum** of S, and we write

$$\beta = \sup S.$$

Example

The Completeness Axiom

It is one thing to define an object and another to show that it always exists. For example:

- Does it make sense to define the smallest positive real number?
- Does the empty set, as a subset of $\mathbb{R}$ have a supremum?

Since the property that a subset of $\mathbb{R}$ has a supremum, although it might not belong to the set, is indispensable in the rigorous development of calculus, we take it as an axiom for the real numbers.

(i) If a nonempty set of real numbers is bounded above, then it has a supremum.

This property is called **completeness**, and we say that $\mathbb{R}$ together with the binary operations of addition and multiplication, and the order relation $<$ is a complete ordered field.

In fact it can be shown that the real number system is essentially the only complete ordered field.

Theorem

If a nonempty set S of real numbers is bounded above, then $\sup S$ is the unique real number β such that

(1) $x \leq \beta$ for all x in S, and

(2) if $\epsilon > 0$, no matter how small, there exists an x_0 in S with $x_0 > \beta - \epsilon$.

Some notation

We will often define a set S by writing

$$S = \{x \mid \cdots\},$$

which means that S consists of all x that satisfy the conditions to the right of the vertical bar.

For instance, if we let S denote the set of negative real numbers, we might write

$$S = \{x \in \mathbb{R} \mid x < 0\},$$

and if we let S_1 denote the set of negative integers we could write

$$\begin{aligned} S_1 &= \{x \in \mathbb{Z} \mid x < 0 \text{ and } x \text{ is an integer}\} \\ &= \{x \in \mathbb{Z} \mid x \text{ is a negative integer}\} \\ &= \{x \in \mathbb{Z} \mid x < 0\} \end{aligned}$$

The Archimedean Property

The property of the real numbers described in the next theorem is called the **Archimedean property**.

Intuitively, it states that it is possible to exceed any positive number, no matter how large, by adding an arbitrary positive number, no matter how small, to itself sufficiently many times.

This is an important property, as it

- ensures every real number can be approximated arbitrarily well by rationals, and
- allows us to rigorously define what we mean by a limit, and hence integration and differentiation.

Theorem (Archimedean Property)
If ρ and ϵ are positive real numbers, then there exists a natural number n such that $n\epsilon > \rho$.

Density of the Rationals and Irrationals

Definition

A set D is **dense in the reals** if every open interval $(a, b) \subset \mathbb{R}$ contains an element of D.

Theorem

The rational numbers are dense in the reals, that is, if a and b are real numbers with $a < b$, then there is a rational number p/q such that $a < p/q < b$.

Example

The rational number system is not complete.

Theorem

The set of irrational numbers are dense in the reals, that is, if a and b are real numbers with $a < b$, then there is a irrational number t such that $a < t < b$.

Infimum of a set

A set S of real numbers is **bounded below** if there is a real number a with $x \geq a$ whenever $x \in S$.

In this case, a is a **lower bound** of S. If a is a lower bound of S, so is any smaller number, because of property of transitivity of our order relation $<$.

If α is a lower bound of S, but no number greater than α is, then α is an **infimum** of S, and we write

$$\alpha = \inf S.$$

Geometrically, this means that there are no points of S to the left of α, but there is at least one point of S to the left of any number greater than α.

Theorem

If a nonempty set S of real numbers is bounded below, then $\inf S$ is the unique real number α such that

(a) $x \geq \alpha$ for all x in S, and

(b) if $\epsilon > 0$, no matter how small, there exists an x_0 in S with $x_0 < \alpha + \epsilon$.

The extended real number system

A nonempty set S of real numbers is **unbounded above** if it has no upper bound, and **unbounded below** if it has no lower bound.

It is convenient to adjoin to the real number system two fictitious points, $+\infty$, which we usually write more simply as ∞, and $-\infty$, and to define the order relationships between them and any real number x by

$$-\infty < x < \infty.$$

We call ∞ and $-\infty$ **points at infinity**.

If S is a nonempty set of reals, we write

$$\sup S = \infty$$

to indicate that S is unbounded above, and

$$\inf S = -\infty$$

to indicate that S is unbounded below.

Example

The real number system with ∞ and $-\infty$ adjoined is called the **extended real number system**, or simply the *extended reals*.

A member of the extended reals differing from $-\infty$ and ∞ is called **finite**, that is, an ordinary real number is finite.

The arithmetic relationships among ∞, $-\infty$, and the real numbers are defined as follows.

(a) If a is any real number, then

$$\begin{aligned} a + \infty &= \quad \infty + a = \quad \infty, \\ a - \infty &= -\infty + a = -\infty, \\ \frac{a}{\infty} &= \frac{a}{-\infty} = 0. \end{aligned}$$

(b) If $a > 0$, then

$$a \cdot \infty = \infty \cdot a = \infty, \quad \text{and} \quad a \cdot (-\infty) = (-\infty) \cdot a = -\infty.$$

(c) If $a < 0$, then

$$a \cdot \infty = \infty \cdot a = -\infty \quad \text{and} \quad a \cdot (-\infty) = (-\infty) \cdot a = \infty.$$

We also define

$$\infty + \infty = \infty \cdot \infty = (-\infty) \cdot (-\infty) = \infty$$

and

$$-\infty - \infty = \infty \cdot (-\infty) = (-\infty) \cdot \infty = -\infty.$$

Finally, we set

$$|\infty| = |-\infty| = \infty.$$

It is not useful to define $\infty - \infty$, $0 \cdot \infty$, ∞/∞, and $0/0$. These are sometimes called **indeterminate forms**, and left undefined.

Some set theory and the topology of $\mathbb{R}$

The following definition applies to arbitrary sets, with the understanding that the members of all sets under consideration in any given context come from a specific collection of elements, called the **universal set**. For most of this course, the universal set is the real numbers.

Definition Let S and T be sets.

(a) We say that S **contains** T, and write $S \supset T$ or $T \subset S$, if every member of T is also in S. In this case, T is called a **subset** of S.

(b) We denote by $S - T$ the set of elements that are in S but not in T.

(c) We say that S **equals** T, and we write $S = T$, if S contains T and T contains S. Thus, $S = T$ if and only if S and T have the same members.

(d) We say that S **strictly contains** T, if S contains T but T does not contain S, that is, if every member of T is also in S, but at least one member of S is not in T.

(e) The **complement** of S, denoted by S^c, is the set of elements in the universal set that are not in S.

(f) The **union** of S and T, denoted by $S \cup T$, is the set of elements in at least one of S and T.

(g) The **intersection** of S and T, denoted by $S \cap T$, is the set of elements in both S and T. If $S \cap T = \emptyset$, the empty set, then S and T are said to be **disjoint sets**.

(h) A set with only one member is called a **singleton set**.

Example


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Every set S contains the empty set $\emptyset$. For if $\emptyset$ is was not contained in S, this means that some member of $\emptyset$ is not in S. However, as $\emptyset$ has no members, this is a cannot happen.

If S is any set, then

$$(S^c)^c = S \quad \text{and} \quad S \cap S^c = \emptyset.$$

If S is a set of real numbers, then $S \cup S^c = \mathbb{R}$.

The definitions of union and intersection have generalisations: If $\mathcal{F}$ is an arbitrary collection of sets, then $\cup\{S \mid S \in \mathcal{F}\}$ is the set of all elements that are members of at least one of the sets in $\mathcal{F}$.

Similarly, $\cap\{S \mid S \in \mathcal{F}\}$ is the set of all elements that are members of every set in $\mathcal{F}$.

The union and intersection of finitely many sets $S_1, \ldots, S_n$ are also written as

$$\bigcup_{k=1}^{n} S_k \quad \text{and} \quad \bigcap_{k=1}^{n} S_k.$$

The union and intersection of an infinite sequence $\{S_k\}_{k=1}^{\infty}$ of sets are written as

$$\bigcup_{k=1}^{\infty} S_k \quad \text{and} \quad \bigcap_{k=1}^{\infty} S_k.$$

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If a and b are in the extended reals and $a < b$, then the **open interval** (a, b) is defined by

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

The open intervals (a, ∞) and $(-\infty, b)$ are **semi-infinite** if a and b are finite, and $(-\infty, \infty)$ is the entire real line $\mathbb{R}$.

Definition

(a) If x_0 is a real number and $\epsilon > 0$, then the open interval $(x_0 - \epsilon, x_0 + \epsilon)$ is called an **ϵ-neighborhood** of x_0.

(b) If a set S contains an ϵ-neighborhood of x_0, then S is called a **neighborhood** of x_0, and x_0 is said to be an **interior point** of S.

(c) The set of interior points of S is the **interior** of S, denoted by S^0.

(d) If every point of S is an interior point (that is, $S^0 = S$), then S is said to be **open**.

(e) A set S is **closed** if S^c is open.

The idea of a neighborhood is fundamental and occurs in many other contexts. Whatever the context, the idea is the same: some definition of **closeness** is given (for example, two real numbers are **close** if their difference is **small**), and a neighborhood of a point x_0 is a set that contains all points sufficiently close to x_0.

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Definition

A **deleted neighborhood** of a point x_0 is a set that contains every point of some neighborhood of x_0 except for x_0 itself.

Theorem

(a) An arbitrary union of open sets is open.

(b) An arbitrary intersection of closed sets is closed.

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Definition Let S be a subset of $\mathbb{R}$.

(a) A point $x_0 \in \mathbb{R}$ is a **limit point** of S if every deleted neighborhood of x_0 contains a point of S.

(b) A point $x_0 \in \mathbb{R}$ is a **boundary point** of S if every neighborhood of x_0 contains at least one point in S and one not in S. The set of boundary points of S is called the **boundary** of S and is denoted by ∂S, and the **closure** of S, denoted by $\overline{S}$, is defined by $\overline{S} = S \cup \partial S$.

(c) A point $x_0 \in \mathbb{R}$ is an **isolated point** of S if $x_0 \in S$ and there is a neighborhood of x_0 that contains no other point of S.

(d) A point $x_0 \in \mathbb{R}$ is **exterior** to S if x_0 is in the interior of S^c. The collection of such points is the **exterior** of S.

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Theorem

A set S is closed if and only if no point of S^c is a limit point of S.

Corollary

A set is closed if and only if it contains all its limit points.

Open coverings

A collection $\mathcal{H}$ of open sets is an **open covering** of a set S if every point in S is contained in a set H belonging to $\mathcal{H}$, that is,

$$S \subset \cup\{H \mid H \in \mathcal{H}\}.$$

Example

Theorem (Heine-Borel Theorem)

If $\mathcal{H}$ is an open covering of a closed and bounded subset S of the real line, then S has an open covering $\widetilde{\mathcal{H}}$ consisting of finitely many open sets belonging to $\mathcal{H}$.

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Theorem (Bolzano-Weierstrass Theorem)

Every bounded infinite set of real numbers has at least one limit point.

2 – Differential Calculus of Functions

In this chapter we study the differential calculus of functions of one variable.

- § 2.1 introduces the concept of function and discusses arithmetic operations on functions, limits, one-sided limits, limits at $\pm\infty$, and monotonic functions.
- § 2.2 defines continuity and discusses removable discontinuities, composite functions, bounded functions, the intermediate value theorem, uniform continuity, and additional properties of monotonic functions.
- § 2.3 introduces the derivative and its geometric interpretation. Topics covered include the interchange of differentiation and arithmetic operations, the chain rule, one-sided derivatives, extreme values of a differentiable function, Rolle's theorem, the intermediate value theorem for derivatives, and the mean value theorem and its consequences.

2.1 Functions and Limits

Functions

- A rule f that assigns to each member of a nonempty set D a unique member of a set Y is a **function from D to Y**.
- We write the relationship between a member x of D and the member y of Y that f assigns to x as

$$y = f(x).$$

- The set D is called the **domain** of f, denoted by D_f.
- The members of Y are the possible **values** of f.
- If $y_0 \in Y$ and there is an x_0 in D such that $f(x_0) = y_0$ then we say that f **attains**, or **assumes**, the value y_0.
- The set of values attained by f is the **range** of f.
- A **real-valued function of a real variable** is a function whose domain and range are both subsets of the reals.

Example

Definition

If $D_f \cap D_g \neq \emptyset$, then $f+g$, $f-g$, and $f \cdot g$ are defined on $D_f \cap D_g$ by

$$\begin{aligned}(f+g)(x) &= f(x)+g(x),\\ (f-g)(x) &= f(x)-g(x),\\ (f\cdot g)(x) &= f(x)\cdot g(x).\end{aligned}$$

The quotient f/g also is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

for x in $D_f \cap D_g$ such that $g(x) \neq 0$.

Example

Limits

Definition

We say that $f(x)$ **approaches the limit** L **as** x **approaches** x_0, and write

$$\lim_{x \to x_0} f(x) = L,$$

if f is defined on some deleted neighbourhood of x_0 and, for every $\epsilon > 0$, there is a $\delta > 0$ such that if $|x - x_0| < \delta$, then

$$|f(x) - L| < \epsilon.$$

Example